

Competitive Programming

Section 65: Polynomial Algorithms And Recurrences

Competitive Programming Course

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1 Why polynomials appear in contests

A coefficient array can encode choices. If $A[i]$ counts ways to make sum i and $B[j]$ counts ways to make sum j , then the coefficient of x^s in their product is

$$C[s] = \sum(A[i] * B[s-i]).$$

That is exactly the number of ways to choose one object from each family with total s . Polynomial multiplication is therefore a fast engine for many-sum DP, correlations, string alignment, and linear recurrences.

2 Convolution and correlation are different index equations

Convolution uses $i+j=s$. Correlation uses a difference such as $j-i=r$. Reverse one array to convert a difference into a sum. Never memorize only an offset: derive the requested indices on paper.

For cyclic comparison

```
score[r] = sum_i [s[i] = t[(i+r) mod n]],
```

duplicate the indicator array of t , reverse the indicator of s , and read coefficient $n-1+r$.

```
reverse_s[n-1-i] pairs with doubled_t[r+i]  
indices add to n-1+r.
```

3 The NTT model

Evaluating a degree- n polynomial at sufficiently many points, multiplying pointwise, and interpolating recovers its product. The number theoretic transform performs these operations inside a finite field.

We use $\text{MOD} = 998244353 = 119 \cdot 2^{23} + 1$ and primitive root 3. For every power-of-two length up to 2^{23} , the field contains a root w whose powers cycle through exactly that length.

At one butterfly:

```
u = left
v = right * current_root
new_left  = u + v
new_right = u - v
```

Each stage doubles the block length. Bit-reversal places coefficients in the order required by the iterative butterfly graph. The inverse transform uses inverse roots and finally divides every value by the transform length.

3.1 A four-position trace

Ignoring the concrete modular root, the dataflow is:

```
(a0, a1, a2, a3)
bit reverse -> (a0, a2, a1, a3)
length 2    -> (a0+a2, a0-a2, a1+a3, a1-a3)
length 4    -> combine the two halves with powers 1 and w
```

Every stage performs linear work, and there are $\log n$ stages. Forward and inverse transforms are $O(n \log n)$.

4 Product trees for many factors

Multiplying factors from left to right can repeatedly grow a large polynomial. Instead, multiply balanced halves recursively (or always combine the two smallest polynomials). Truncate every intermediate result after the largest coefficient the problem can query.

For an item contributing any value from 0 through a , its generating polynomial is

$$1 + x + x^2 + \dots + x^a.$$

The coefficient of the product counts bounded compositions. This is a direct application of the ordinary generating-function product rule, not a symbolic trick.

5 Huge indices in linear recurrences

Suppose

$$a[m] = c[0]a[m-1] + c[1]a[m-2] + \dots + c[k-1]a[m-k].$$

Introduce the characteristic relation

$$x^k = c[0]x^{(k-1)} + c[1]x^{(k-2)} + \dots + c[k-1].$$

Reduce x^n modulo this relation:

$$x^n = p[0] + p[1]x + \dots + p[k-1]x^{(k-1)}.$$

The same relation holds for every shifted recurrence window, so

$$a[n] = p[0]a[0] + \dots + p[k-1]a[k-1].$$

Binary exponentiation computes the reduced polynomial x^n . Multiplication followed by descending-degree reduction costs $O(k^2)$; total cost is $O(k^2 \log n)$, independent of the magnitude of n except for its bits.

6 Discovering a recurrence from observations

Berlekamp–Massey maintains the shortest relation that explains the processed prefix. When the current term disagrees with that relation, the discrepancy tells us how much of an older successful relation must be shifted and scaled to repair the new position.

The important conceptual states are:

- **C**: current relation;
- **B**: relation saved at the last length increase;
- **b**: discrepancy at that save point;
- **shift**: how far the saved relation must move;
- **L**: current recurrence length.

If the discrepancy is zero, the relation already predicts this term. If it is nonzero, subtract **discrepancy** / **b** times the shifted saved relation. When the processed prefix has become too long for the previous **L**, promote the old current relation to the saved one and increase **L**.

The theorem requires a field, because discrepancy scaling divides by **b**. Our prime modulus provides inverses for every nonzero **b**.

7 Recognition table

Statement shape	Algebraic model
all pair sums or matching shifts	convolution/correlation
combine many independent bounded choices	product of generating polynomials
recurrence order moderate, index enormous	polynomial reduction / Kitamasa
many observed terms, recurrence hidden	Berlekamp–Massey, then recurrence evaluation

8 Common mistakes

- Forgetting that linear convolution needs padding to at least $\text{len}(A) + \text{len}(B) - 1$.
- Reading the wrong correlation coefficient because the reversal was guessed.
- Using an NTT length beyond the power of two supported by the modulus.
- Reducing a high degree upward; reduction must go from high degree downward.
- Running Berlekamp–Massey over a composite modulus as if every discrepancy were invertible.
- Assuming a short observed prefix uniquely determines the future without the guarantee in the statement.

9 Exercises

A isolates correlation indexing. B forces the generating-function product model. C separates recurrence evaluation from matrix exponentiation. D is a blind synthesis problem: the lesson names the machinery but does not reveal how its input should be normalized or checked.

10 Before submitting

- Test convolution against $O(nm)$ on random arrays of length at most eight.
- Test cyclic shifts 0, 1, and $n-1$ with a hand-written pair.
- For recurrence code, test $n < k$, $k=1$, all-zero coefficients, and Fibonacci.
- Feed Berlekamp–Massey a constant sequence, a geometric sequence, and leading zeros before a nonzero recurrence.